Started on	Wednesday, 26 August 2026, 9:58 PM
State	Finished
Completed on	Wednesday, 26 August 2026, 10:04 PM
Time taken	5 mins 58 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

Reset answer

```
#include<stdio.h>
int main(){
    int n;
    int c=0;
    if(scanf("%d",&n)!=1){
        return 0;
    }
    c++;
    int i=n/2;
    while(c++,i<n){
        int j=1;
        while(c++,j<n){
```

```
            int k=1;
            while(c++,k<n){
                c++;
                k=k*2;
            }
            j=2*j;
        }
        i++;
    }
    printf("%d",c);
    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.